

Title: AI BlogNest API.

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Description:

The **ContentGen** backend is a powerful, secure, and intelligent REST API platform designed to simplify blog creation, content management, and AI-assisted writing. Built with **Node.js** and **Express.js**, the application uses **MongoDB** as its primary database with **Mongoose ODM** for efficient data modeling, validation, and seamless database interactions.

The platform integrates **Google Gemini AI** to provide intelligent features such as **AI-powered blog content generation**, **AI FAQ assistance**, **AI WeatherWise insights**, and **AI FitTrack**

recommendations, enabling users to create high-quality, context-aware content with minimal effort.

Security is a core aspect of the architecture, implementing **JWT (JSON Web Tokens)** for stateless authentication, **role-based access control (Admin, Editor, Author, Reader)** for authorization, and **bcrypt.js** for secure password hashing. The backend also supports comprehensive blog management, categories and tags, comment management, media metadata handling, advanced search capabilities, and centralized error handling. Additional security mechanisms such as request validation, input sanitization, rate limiting, and structured logging ensure the platform remains reliable, scalable, and resilient against common web vulnerabilities.

Designed with a modular and scalable architecture, **ContentGen** serves as a modern AI-powered Content Management System (CMS) that combines traditional blogging capabilities with Generative AI to deliver a smarter, faster, and more personalized content creation experience.

Scenario-Based Case Study:

Background

Priya is a content creator who manages multiple blogs but struggles to organize posts, schedule publications, manage categories/tags, and track reader engagement efficiently.

Problem

- Difficulty managing multiple drafts, categories, and tags.
- Limited visibility into post performance and engagement analytics.
- Time-consuming processes for content approval and comment moderation.
- No streamlined system for handling media uploads or metadata.
- Lack of collaborative permission tiers for editors, authors, and administrators.

Solution

The BlogNest backend delivers a centralized, secure API wrapper that decouples application business logic from frontend delivery systems. It introduces an internal permission matrix controlling specialized services for content scheduling, category taxonomy, media path registration, and granular user roles.

Usage

- **Author Usage:** Creates blog posts, uploads media pathways, organizes content drafts, and reviews their distinct post-performance data.
- **Editor/Admin Usage:** Accesses protected endpoints to review content queues, change publishing statuses, moderate visitor comments, and orchestrate user privilege upgrades.

- **Reader Usage:** Queries public list routes to consume posts and invokes comment creation endpoints.

Outcome

By offloading workflow controls to a robust backend, content execution delays are minimized. Data structural integrity is strictly maintained through schema validation, security filters guarantee that only authorized roles perform administrative shifts, and persistent tracking provides raw analytics to measure post optimization metrics accurately.

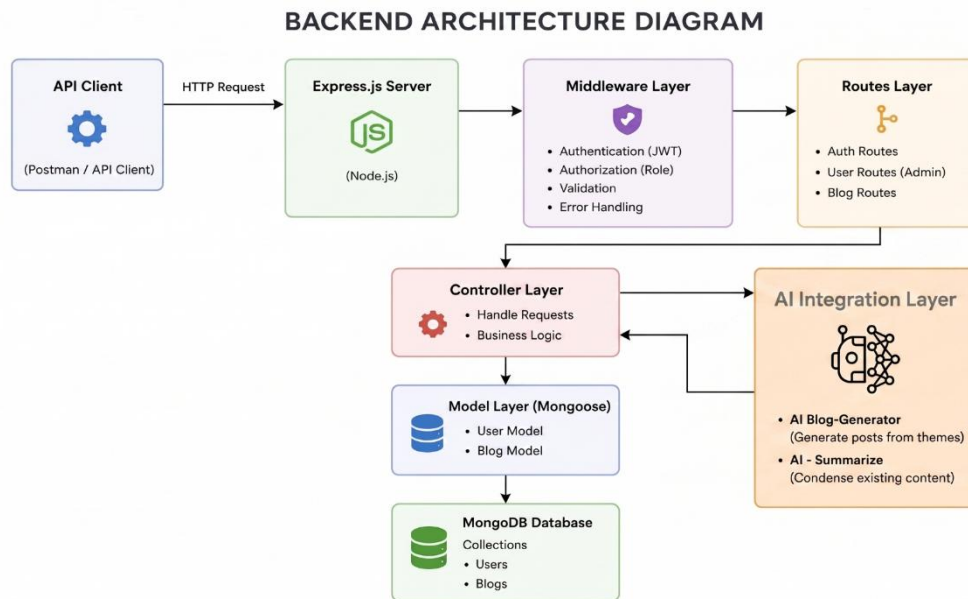
1. Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux (supports cross-platform operations).
- **Node.js (v16 or above):** Runtime ecosystem running server logic and routing infrastructure.
- **npm (v8 or above):** Node package manager to control operational dependencies.
- **Express.js:** Lightweight routing web framework to construct backend RESTful entry points.
- **MongoDB:** Document NoSQL engine storing users, posts, categories, comments, and analytics metrics.
- **Postman:** API verification toolkit to assert schema validation responses across administrative route shields.
- **Code Editor:** Visual Studio Code or similar IDE environment.

2. Hardware Requirements

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- **RAM:** 8 GB minimum (16 GB recommended for concurrent instances of MongoDB, processing layers, and execution nodes).
- **Storage:** 1 GB of available disk workspace.

Epic:-1 PROJECT ARCHITECTURE :



The BlogNest application exposes structured endpoints handled by a layered server architecture model. Incoming requests undergo filter processing before interacting with data services.

Components Definition

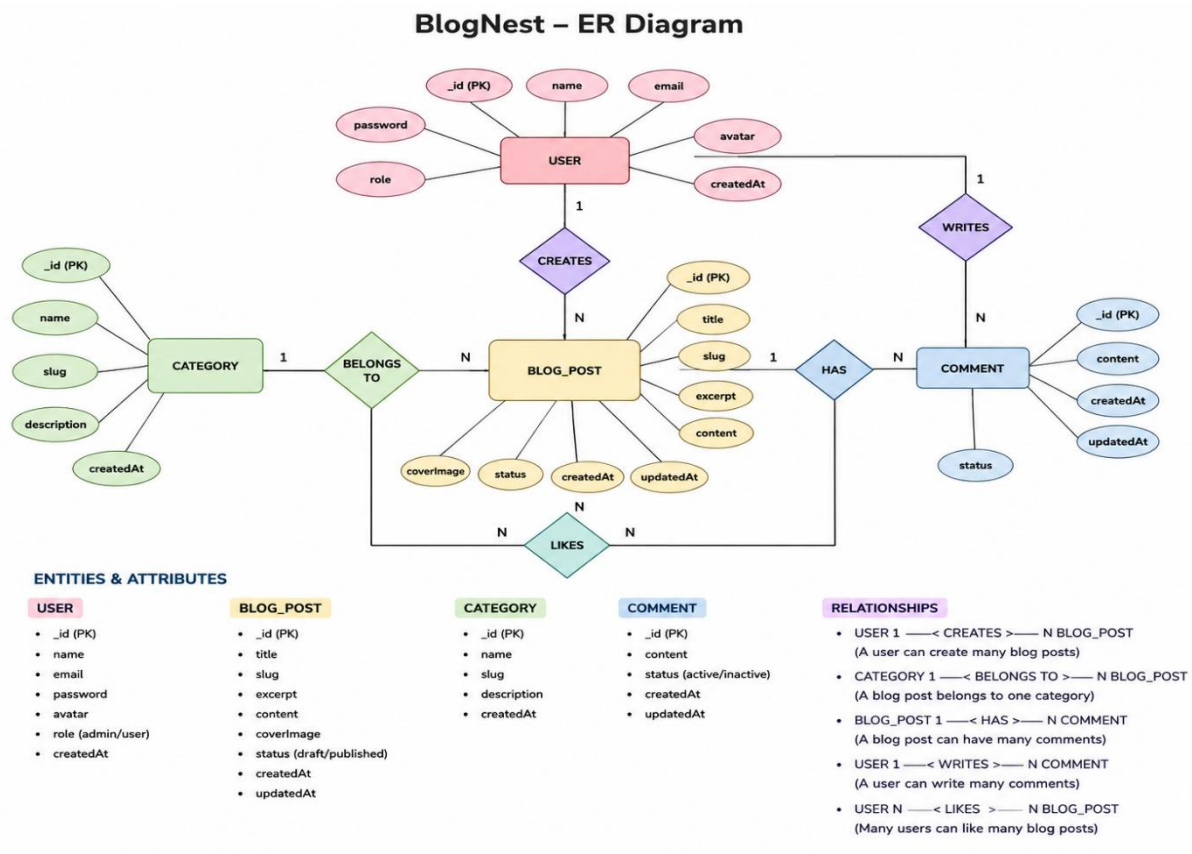
- **Interface:** API tools like Postman query distinct route paths to extract JSON entities.
- **Express Server Gateway:** Monitors network ports, maps explicit HTTP methods (GET, POST, PUT, DELETE), parses JSON payloads, and handles CORS headers.
- **Authentication & Role Matrix Middleware:** Extracts tokens from headers, validates user context signatures, and evaluates if the specific role satisfies authorization thresholds for guarded resource paths.
- **Granular Services (Logic Modules):** Enforces isolated domain rules for blog lifecycles (Draft, Pending Approval, Scheduled, Published), comment queues, and system reporting indicators.
- **Database Interface Layer:** Mongoose translates operational application logic into performant native MongoDB queries.
- **AI Flow (Service Integration):** Provides intelligent content automation, including automated blog post generation and real-time content summarization. This layer connects to external AI models via specialized endpoints (/api/ai/generate-blog and

/api/ai/summarize), allowing the controller layer to offload complex text processing tasks and enhance content creation workflows.

- **BlogGenerator with AI:** A specialized service that automates the generation of blog content, which integrates with the AI Flow service to streamline the drafting process.

Entity-Relationship (ER) Diagram Description

The underlying document structure manages several collections bound by logical application references:



1. User Entity

- `_id` (Primary Key - ObjectId)
- name (String, Required)
- email (String, Required, Unique)
- password (String, Hashed, Required)
- role (String, Default: "reader")

2. Post Entity

- `_id` (Primary Key - ObjectId)
- `userID` (Foreign Key referencing User Collection)
- `name` (String, Required - represents Title)
- `photo` (Object/String pathway link)
- `message` (String - Core content body)
- `likes` (Number, Default: 0)
- `timestamps` (True)

Key Structural Mappings

- **User to Post (One-to-Many):** An author can submit multiple blog documents.
- **User to Comment (One-to-Many):** Readers/Users can append multiple response comments across independent post nodes.
- **Post to Comment (One-to-Many):** A single blog document stores individual referenced comment submissions.

Key Features

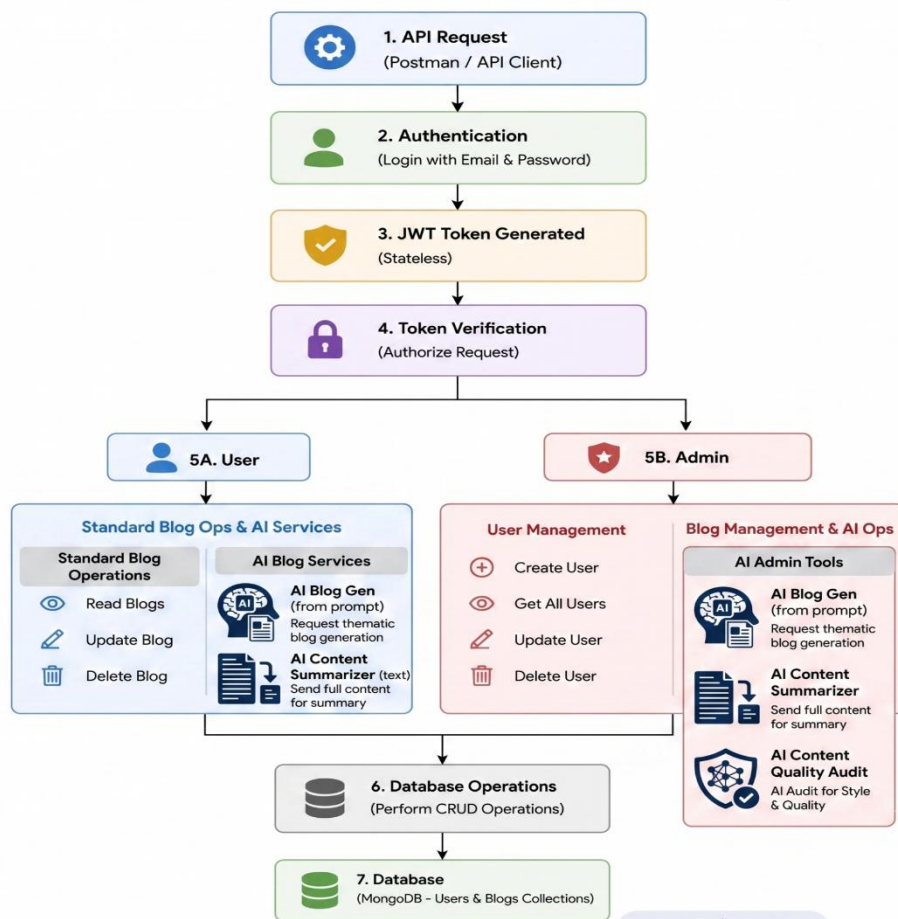
- **Role-Based Route Guarding:** Strict token processing blocks unauthorized tiers from modifying core settings.
- **CRUD Blog Execution Engine:** Complete schema management over content definitions, status states, and asset assignment paths.
- **Categorization System:** Native tag indices facilitate filtering and querying for enhanced discoverability.
- **Comment Moderation Logic:** Protected entry points for updating moderation flags (approved, pending, spam).
- **Sanitization Safeguards:** Payload parameter sanitation prevents XSS vectors and NoSQL parameter manipulation injections.
- **I-Driven Blog Generation Pipeline:** Integrates external Large Language Model (LLM) abstractions and context-aware prompt templates to synthesize full-length, structured blog posts from high-level thematic inputs.
- **Intelligent Content Summarization Engine:** Leverages semantic compression layers to parse extensive article payloads, dynamically generating concise abstracts and automated metadata summaries.

Roles and Responsibilities

- **Admin:** Complete environment override control. Can manage user database registrations, alter role assignments, prune malicious comments, and inspect full logging entries.
- **Editor:** Focuses on editorial workflows. Reviews pending draft posts, marks records for instant publication, and handles category cleanup scripts.
- **Author:** Restricted account access. Can issue creation statements only for posts bound to their explicit account record.
- **Reader:** Consumer access profile. Reads published materials and posts comments under an ongoing authenticated identity.

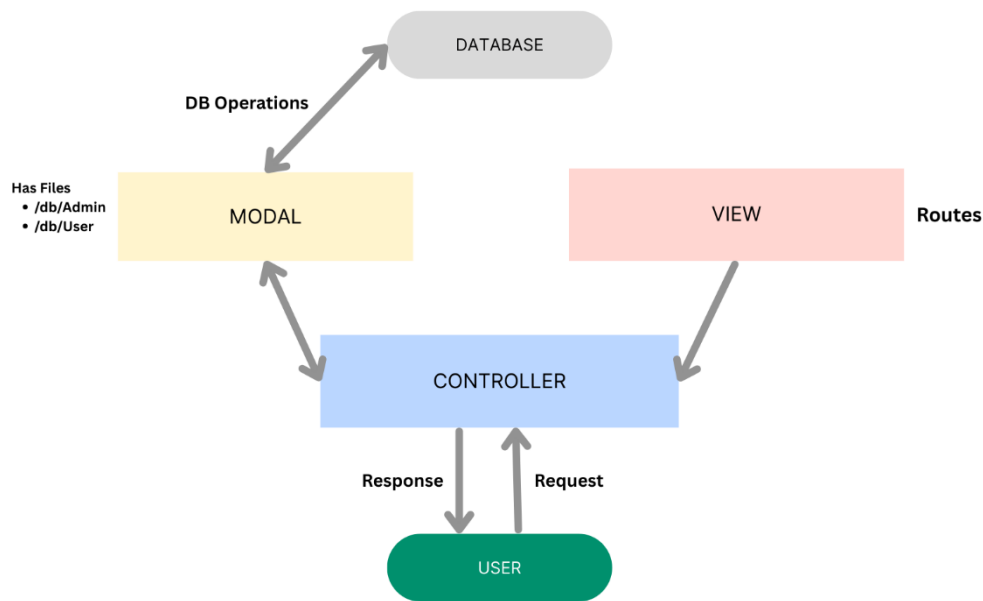
User Flow:

AI-BLOG USER FLOW DIAGRAM (Backend Only)



MVC Architecture Pattern

The system architecture structures its source configuration layers around the Model-View-Controller design pattern to completely decouple components:



Model Layer (Data Blueprint)

Defines structure blueprints via strongly enforced Mongoose schemas mapping explicit fields into raw collection items inside MongoDB. It sets default states, indexing rules, and transformation parameters before validation persistence occurs.

Controller Layer (Execution Matrix)

The intermediary brain of the ecosystem. It captures request vectors from the execution routes, checks structural request constraints, delegates processing workflows directly to models, and packages the resultant output into standard JSON return arrays.

View Layer (Routing Network Interface)

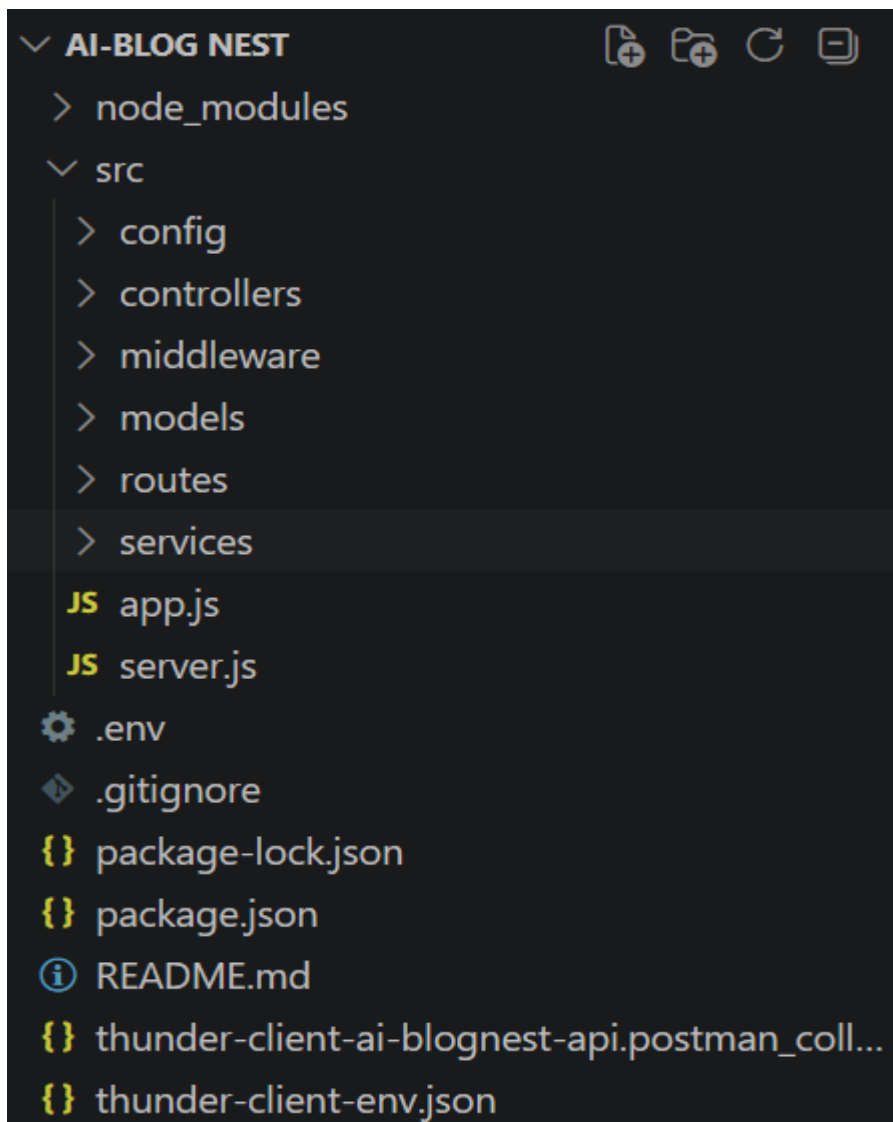
In this pure headless backend service context, standard user-interface template views do not exist. Instead, the interface layer lives as an API **Routing Layer**. It links network requests on specified paths directly to their associated controller handler logic maps.

Epic2: Project Setup and Configuration

Creating project folder

1. Create a new folder with your <project name>.
2. Inside that folder create one new folders.
3. Name another one as Server.

4. Now open that folder in VS Code.



Server setup (npm init)

- Open Server folder in terminal of VScode.
 - `npm init -y`
- Create files:
 - `server.js`
- Create folders:
 - `models`

- controllers
- routes

Epic 3: Backend Development:

Server.js

```
JS server.js X
src > JS server.js > ...
1  const dotenv = require('dotenv');
2  const app = require('./app');
3  const connectDB = require('./config/db');
4
5  dotenv.config();
6  connectDB();
7
8  const PORT = process.env.PORT || 5000;
9  app.listen(PORT, () => {
10 |   console.log(`Server running on port ${PORT}`);
11 | });
12
```

App.js

```
JS app.js X
src > JS app.js > ...
1  const express = require('express');
2  const cors = require('cors');
3  const morgan = require('morgan');
4  const authRoutes = require('./routes/authRoutes');
5  const blogRoutes = require('./routes/blogRoutes');
6  const aiRoutes = require('./routes/aiRoutes');
7  const errorMiddleware = require('./middleware/errorMiddleware');
8
9  const app = express();
10
11  app.use(cors());
12  app.use(express.json());
13  app.use(morgan('dev'));
14
15  app.use('/api/auth', authRoutes);
16  app.use('/api/blogs', blogRoutes);
17  app.use('/api/ai', aiRoutes);
18
19  app.use(errorMiddleware);
20
21  app.get('/', (req, res) => {
22 |   res.json({ message: 'AI BlogNest API is running' });
23 | });
24
25  module.exports = app;
26
```

Epic4: Database Configuration

DB Connection:-

```
Explorer (Ctrl+Shift+E)
src > config > JS dbjs > ...
1  const mongoose = require('mongoose');
2
3  const connectDB = async () => {
4    const mongoUri = process.env.MONGO_URI;
5    if (!mongoUri) {
6      console.error('MongoDB connection error: MONGO_URI is not defined in environment variables');
7      process.exit(1);
8    }
9
10   try {
11     const conn = await mongoose.connect(mongoUri, {
12       useNewUrlParser: true,
13       useUnifiedTopology: true,
14     });
15     console.log(`MongoDB connected: ${conn.connection.host}`);
16   } catch (error) {
17     console.error('MongoDB connection error: ${error.message}`);
18     process.exit(1);
19   }
20 };
21
22 module.exports = connectDB;
```

1. Configure MongoDB:

- Install Mongoose.

In your Node.js project directory, run:

`npm install mongoose`

- Create database connection.
- Create Schemas & Models.

Create Database connection:

Now, make sure the database is connected before performing any of the actions through the backend. The connection code looks similar to the one provided below

Mongoose Schema Definitions

1. Identity Resource Document Schema (schemas/userModel.js)

Enforces field criteria validations on incoming database creations:

```
JS User.js  X
src > models > JS User.js > ...
1  const mongoose = require('mongoose');
2
3  const userSchema = new mongoose.Schema(
4  {
5      name: {
6          type: String,
7          required: [true, 'Name is required'],
8          trim: true,
9      },
10     email: {
11         type: String,
12         required: [true, 'Email is required'],
13         unique: true,
14         lowercase: true,
15         trim: true,
16     },
17     password: {
18         type: String,
19         required: [true, 'Password is required'],
20         minlength: 6,
21     },
22 },
23 {
24     timestamps: true,
25 }
26 );
27
28 module.exports = mongoose.model('User', userSchema);
29
```

Epic 5: Project Executions:

- Install Postman or ThunderClient (VS Code extension).
- Start the backend server:

“npm start”

API Testing

User Registration :

The screenshot shows the Postman API client interface. The top bar displays the URL `http://localhost:5000/api/auth/register` and the method `POST`. The request body is a JSON object:

```
{  "name": "theja",  "email": "theja@gmail.com",  "password": "123456"}
```

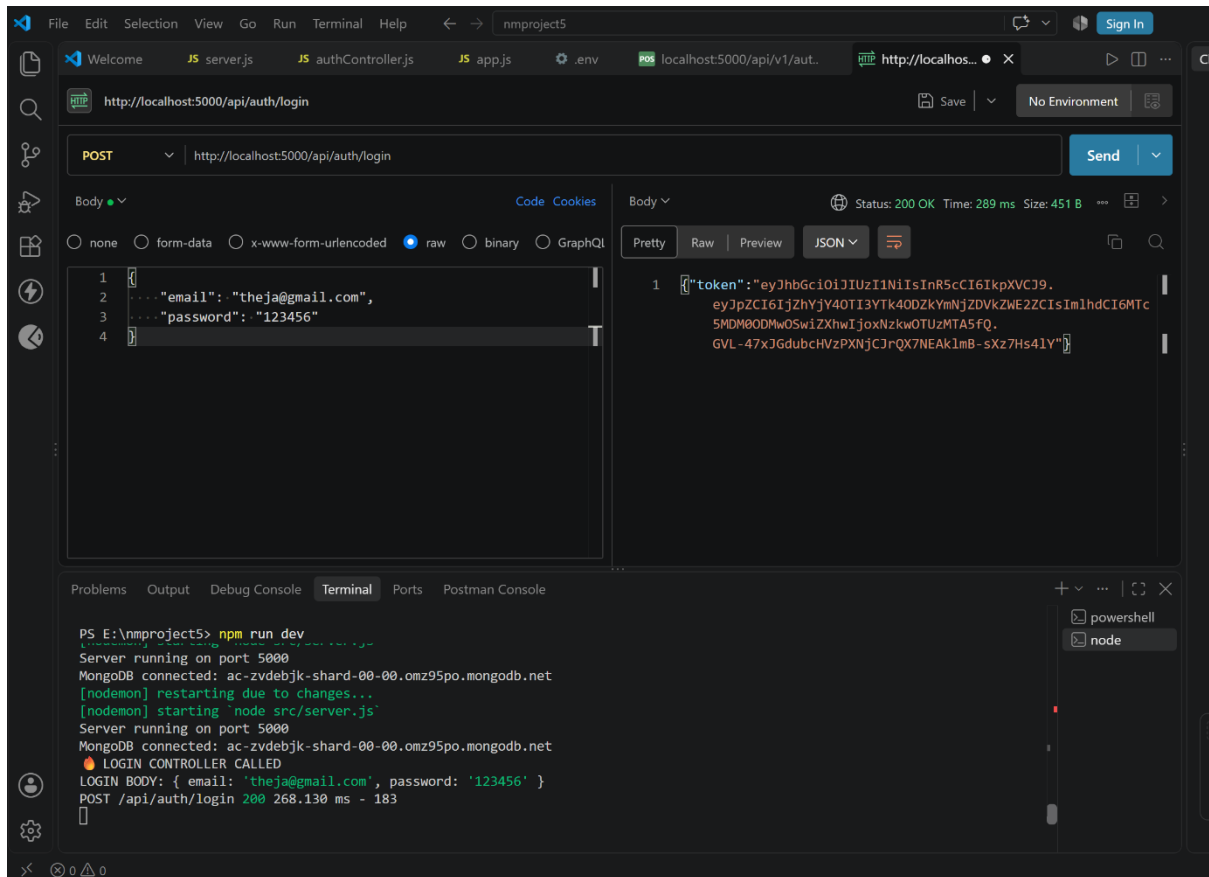
The response status is `400 Bad Request` with a time of `68 ms` and a size of `314 B`. The response body is a JSON object:

```
{  "message": "Email already registered"}
```

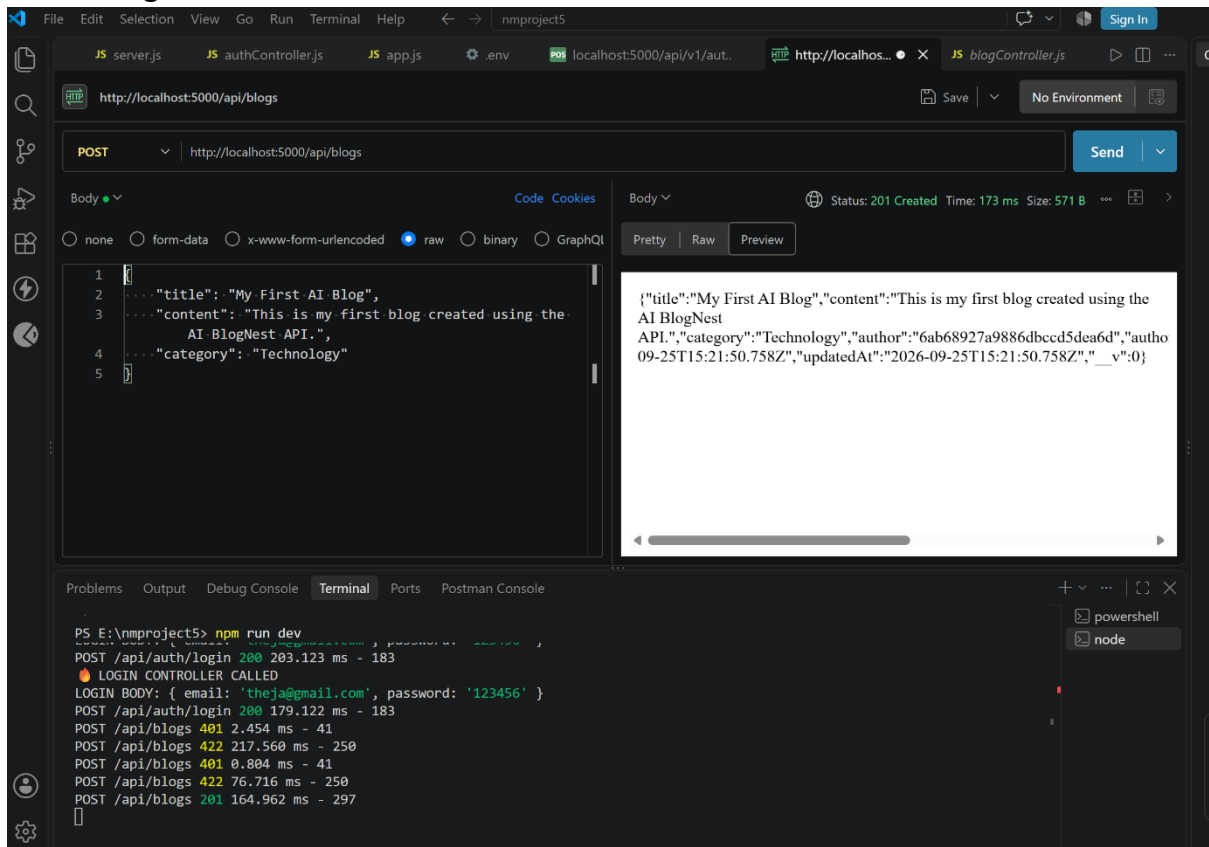
The bottom panel shows the terminal output of the application running on port 5000. The output indicates that the login controller was called and the login request was successful, but the registration request failed with a 400 status code.

```
PS E:\nmp\project5> npm run dev
[nodemon] restarting due to changes...
[nodemon] starting 'node src/server.js'
Server running on port 5000
MongoDB connected: ac-zvdebjk-shard-00-00.omz95po.mongodb.net
LOGIN CONTROLLER CALLED
LOGIN BODY: { email: 'theja@gmail.com', password: '123456' }
POST /api/auth/login 200 268.130 ms - 183
BODY: { name: 'theja', email: 'theja@gmail.com', password: '123456' }
POST /api/auth/register 400 60.918 ms - 38
```

User Login:



Create Blogs :



Fetch Blogs :

The screenshot displays the VS Code editor with a REST client request and its response, along with a terminal window showing the command-line execution of the application.

REST Client Request:

```
GET http://localhost:5000/api/blogs
```

Request Body (raw):

```
1 {
2   ... "title": "My First AI Blog",
3   ... "content": "This is my first blog created using the
      AI BlogNest API.",
4   ... "category": "Technology"
5 }
```

Response Body (raw):

```
{ "_id": "6ab6918e2d6c17563dda3358", "title": "My First AI
Blog", "content": "This is my first blog created using the AI BlogNest
API.", "category": "Technology", "author":
{ " _id": "6ab68927a9886dbccd5dea6d", "name": "Theja", "email": "theja@gmail.co
09-25T15:21:50.758Z", "updatedAt": "2026-09-25T15:21:50.758Z", "__v": 0 } }
```

Terminal Output:

```
PS_E:\nmproject5> npm run dev
... LOGIN CONTROLLER CALLED
LOGIN BODY: { email: 'theja@gmail.com', password: '123456' }
POST /api/auth/login 200 179.122 ms - 183
POST /api/blogs 401 2.454 ms - 41
POST /api/blogs 422 217.560 ms - 250
POST /api/blogs 401 0.804 ms - 41
POST /api/blogs 422 76.716 ms - 250
POST /api/blogs 201 164.962 ms - 297
GET /api/blogs 200 284.956 ms - 348
```

Generate AI blogs

POST http://localhost:5000/api/ai/generate-blog

Body: raw

```
1 {
2   "topic": "Node.js",
3   "category": "Technology"
4 }
```

Status: 404 Not Found Time: 1098 ms Size: 710 B

Body: Pretty Raw Preview

```
{\"message\":\"[GoogleGenerativeAI Error]: Error fetching from https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent: [404 Not Found] This model models/gemini-2.5-flash is no longer available to new users. Please update your code to use models/gemini-3.8-flash for the latest features and improvements. We recommend you to use the Interactions API (https://ai.google.dev/gemini-api/docs/get-started).\"}
```

Terminal

```
PS E:\nmproject5> npm run dev
...
at async generateContent (E:\nmproject5\node_modules\@google\generative-ai\dist\index.js:867:22)
at async callGemini (E:\nmproject5\src\services\geminiService.js:18:20)
at async generateBlog (E:\nmproject5\src\controllers\aiController.js:44:20)
POST /api/ai/generate-blog 404 1091.949 ms - 434
```

Summarize Blogs :

POST http://localhost:5000/api/ai/summarize

Body: raw

```
1 {
2   "content": "React.js"
3 }
```

Status: 200 OK Time: 3.51 s Size: 451 B

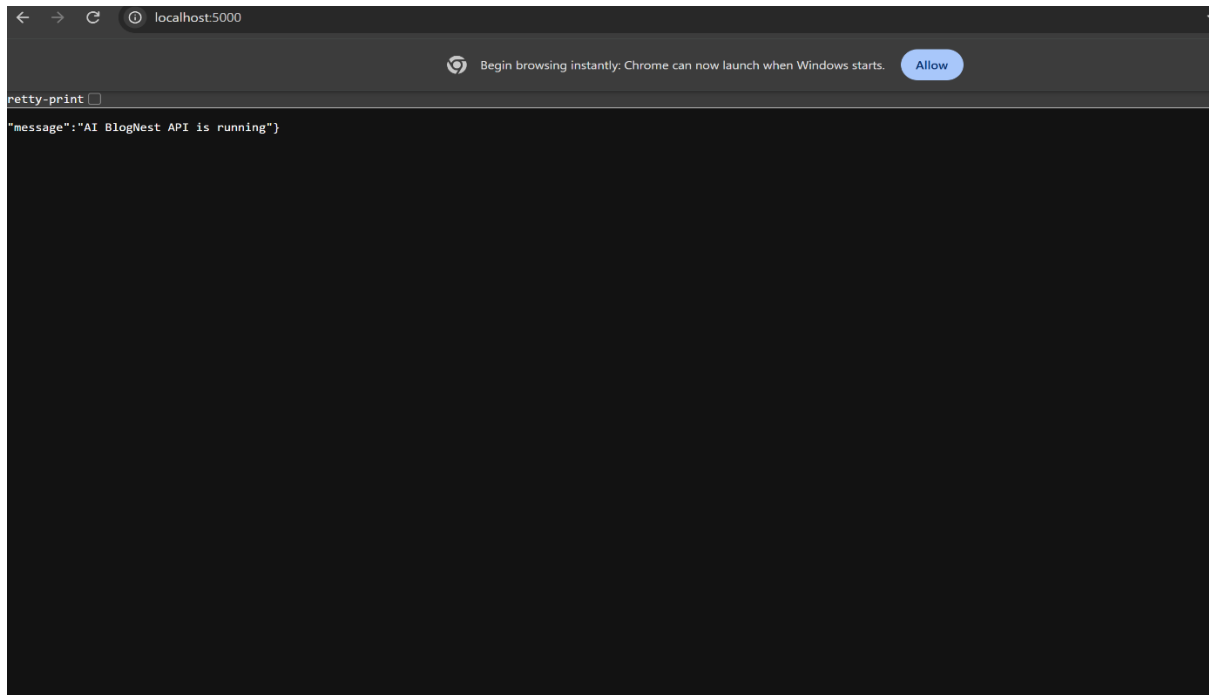
Body: Pretty Raw Preview

```
{\"summary\":\"It looks like you forgot to paste the blog content! Please provide the text about React.js, and I'll be happy to summarize it for you in a short, easy-to-read paragraph.\"}
```

Terminal

```
PS E:\nmproject5> npm run dev
...
[monodemon] starting 'node src/server.js'
Server running on port 5000
MongoDB connected: ac-zvdebjk-shard-00-00.omz95po.mongodb.net
Gemini attempt 1/3
Gemini response received
POST /api/ai/generate-blog 201 2387.032 ms - 1210
Gemini attempt 1/3
Gemini response received
POST /api/ai/summarize 200 3500.124 ms - 183
```

output:



API TESTING VIDEO:

https://drive.google.com/file/d/1x3OXLcDsaqYE787i42tgo2N74_7eXsUP/view?usp=drivesdk

DEMO LINK:

<https://drive.google.com/file/d/1DDzDNRIgZDxd8gZndTO7MjiejqqwULwfM/view?usp=drivesdk>

CodeDrive:

https://drive.google.com/drive/folders/1kbZgb6sU_6zjzPy2-hOuL-bhEo5bFsEQ?usp=sharing